

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar ... taking wif u oni

```
[ ] import pandas as pd

df = pd.read_csv("/content/spam.csv")

df.head()
```

Category		Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Next steps:

Generate code with df

New interactive sheet

```
[ ]  from google.colab import files  
uploaded = files.upload()
```

```
... Choose Files spam.csv
spam.csv(text/csv) - 480130 bytes, last modified: n/a - 100% done
Saving spam.csv to spam.csv
```

```
[ ] df.shape
```

✓ (5572, 2)

```
[ ] df.groupby('Category').describe()
```

Message

```
count unique top
```

freq

Automatic saving failed. This file was updated remotely or in another tab. [Show diff](#)

```
df_ham=df[df['Category']=='ham']
```

```
df_ham_downsample=df_ham.sample(df_spam.shape[0])
```

```
df_ham_downsample.shape
```

(747, 2)

```
df_balanced=pd.concat([df_spam,df_ham_downsample])
```

df_balanced

...			Category	Message	...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...			
5	spam	FreeMsg Hey there darling it's been 3 week's n...			
8	spam	WINNER!! As a valued network customer you have...			
9	spam	Had your mobile 11 months or more? U R entitle...			
11	spam	SIX chances to win CASH! From 100 to 20,000 po...			
...	...	...			
551	ham	Imagine you finally get to sink into that bath...			
5334	ham	Garbage bags, eggs, jam, bread, hannaforw whea...			
806	ham	I dled 3d its very imp			
2052	ham	Hey darlin.. i can pick u up at college if u t...			
3797	ham	They have a thread on the wishlist section of ...			

1494 rows x 2 columns

```
df.groupby('Category').describe()
```

Category	Message			freq
	count	unique	top	
ham	4825	4516	Sorry, I'll call later	30
spam	747	641	Please call our customer service representativ...	4

747/4825

0.15481865284974095

```
df_spam=df[df['Category']=='spam']
```

df_spam

	Category	Message
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
5	spam	FreeMsg Hey there darling it's been 3 week's n...
8	spam	WINNER!! As a valued network customer you have...
9	spam	Had your mobile 11 months or more? U R entitle...
11	spam	SIX chances to win CASH! From 100 to 20,000 po...
...	...	...
5537	spam	Want explicit SEX in 30 secs? Ring 02073162414...
5540	spam	ASKED 3MOBILE IF 0870 CHATLINES INCLU IN FREE ...

df_balanced

	Category	Message	spam
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1
5	spam	FreeMsg Hey there darling it's been 3 week's n...	1
8	spam	WINNER!! As a valued network customer you have...	1
9	spam	Had your mobile 11 months or more? U R entitle...	1
11	spam	SIX chances to win CASH! From 100 to 20,000 po...	1
...	...	...	...
551	ham	Imagine you finally get to sink into that bath...	0
5334	ham	Garbage bags, eggs, jam, bread, hannafor whea...	0
806	ham	I dled 3d its very imp	0
2052	ham	Hey darlin.. i can pick u up at college if u t...	0
3797	ham	They have a thread on the wishlist section of ...	0

1494 rows x 3 columns

Next steps:

Generate code with df_balanced

New interactive sheet

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(df_balanced['Message'],df_balanced['spam'],stratify=df_balanced['spam'])
```

x_test

```
[ ] df_balanced['spam'].value_counts()

count
spam
1    747
0    747

dtype: int64

[ ] bert_preprocess=hub.KerasLayer("https://www.kaggle.com/models/tensorflow/bert/TensorFlow2/en-uncased-preprocess/3")

[ ] bert_encoder=hub.KerasLayer("https://tfhub.dev/tensorflow/bert_en_uncased_L-12_H-768_A-12/4")

[ ] def get_sentence_embedding(sentences):
    preprocessed_text=bert_preprocess(sentences)
    return bert_encoder(preprocessed_text)['pooled_output']

[ ] ▶ get_sentence_embedding([
    "500 discount hurry up!",
    "hey nonsense how are you?"
])

... <tf.Tensor: shape=(2, 768), dtype=float32, numpy=
array([[ -0.8219939 , -0.46519732, -0.8004354 , ..., -0.6150444 ,
        -0.7181412 ,  0.90315425],
       [-0.9373353 , -0.54295063, -0.96882194, ..., -0.8382981 ,
        -0.78021705,  0.9413933 ]], dtype=float32)>

[ ] ▶ e=get_sentence_embedding([
    "ieff bezee"
```

```
get_sentence_embedding([
    "500 discount hurry up!",
    "hey nonsense how are you?"
])
```

```
<tf.Tensor: shape=(2, 768), dtype=float32, numpy=
array([[ -0.8219939 , -0.46519732, -0.8004354 , ..., -0.6150444 ,
        -0.7181412 ,  0.90315425],
       [ -0.9373353 , -0.54295063, -0.96882194, ..., -0.8382981 ,
        -0.78021705,  0.9413933 ]], dtype=float32)>
```

```
e=get_sentence_embedding([
    "jeff bezos",
    "elon musk",
    "bill gates",
    "mango",
    "apple",
    "orange"
])
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
cosine_similarity([e[1]], [e[3]])
```

```
array([[0.89644194]], dtype=float32)
```

```
text_input = tf.keras.layers.Input(shape=(), dtype=tf.string, name='text')
```

```
# Define the BERT layers and model structure to handle symbolic tensors correctly
preprocessed_text=bert_preprocess(text_input)
outputs=bert_encoder(preprocessed_text)
```

```
1 = tf.keras.layers.Dropout(0.1, name="dropout")(outputs['pooled_output'])
```

```
[ ] # Define the BERT layers and model structure to handle symbolic tensors correctly
preprocessed_text=bert_preprocess(text_input)
outputs=bert_encoder(preprocessed_text)
```

```
[ ] l = tf.keras.layers.Dropout(0.1, name="dropout")(outputs['pooled_output'])
l = tf.keras.layers.Dense(1, activation='sigmoid', name='output')(l)
```

WARNING:tensorflow:The following Variables were used in a Lambda layer's call (tf.math.add), but are not present in its tracked objects: <tf.Variable 'out

```
[ ] model = tf.keras.Model(inputs=[text_input], outputs=[l])
model.summary()
```

... Model: "model"

Layer (type)	Output Shape	Param #	Connected to
text (InputLayer)	[(None,)]	0	[]
keras_layer (KerasLayer)	{'input_type_ids': (None, 128), 'input_mask': (None, 128), 'input_word_ids': (None, 128)}	0	['text[0][0]']
keras_layer_1 (KerasLayer)	{'pooled_output': (None, 768), 'sequence_output': (None, 128, 768), 'encoder_outputs': [(None, 128, 768), (None, 128, 768), (None, 128, 768), (None, 128, 768), (None, 128, 768), (None, 128, 768), (None, 128, 768)]	109482241	['keras_layer[0][0]', 'keras_layer[0][1]', 'keras_layer[0][2]']

[]

[]

[]

[]

[]

[]



